Usability Metrics Applied to the EzList System

Translated version of the uploaded report, preserving the original figures and screenshots.

Contents

Usability Metrics Used

Evaluating the usability of interactive systems involves quantitative metrics that reflect the quality of the user's interaction with the system. In this context, ISO/IEC 9241-11 defines usability as "the extent to which a product can be used by specified users to achieve specified goals with effectiveness, efficiency and satisfaction in a specified context of use" (ISO, 2018).

For the analysis of the EzList system, metrics widely adopted in the usability literature were selected, following ISO/IEC 9126-4 guidelines and recommendations proposed by Jakob Nielsen (2001). The metrics considered in this study were:

- Task success rate;
- Average time per task;
- Time-based efficiency;
- Error rate;
- Learning rate;
- Percentage of help requested or required (via video or observation).

1. Task Completion Rate

This metric is directly related to system effectiveness. It indicates the proportion of tasks that users were able to complete successfully, reflecting how well the system enables users to achieve their goals.

Effectiveness = (Number of tasks completed successfully / Total number of tasks performed) × 100%

This index helps identify severe usability failures, such as non-intuitive flows, lack of feedback, or confusing steps. The closer the value is to 100%, the greater the perceived effectiveness of the system.

Example: Suppose 5 users performed 3 tasks each, for a total of 15 tasks. Of these, 13 were completed successfully. Effectiveness = 13 / 15 × 100% = 86.67%. In this example, most tasks were completed successfully, but there is still room to improve interface fluency or the clarity of instructions.

2. Time-Based Efficiency

According to ISO/IEC 9241-11 (ISO, 2018), efficiency is related to the amount of resources used in relation to the accuracy and completeness with which users achieve their goals. The formula used is:

$TBE = [\Sigma\Sigma (n_{ij} / t_{ij})] / (N \times R)$

- n_{ij} = 1 if user j completed task i successfully; 0 otherwise;
- t_{ij} = time spent by user j on task i;
- N = total number of tasks;
- R = total number of users.

In practice, this metric seeks to answer how quickly users are able to complete tasks successfully. The higher the value, the greater the efficiency.

Example: Consider a scenario with 2 users (R = 2) and 2 tasks (N = 2). User 1 completed Task 1 in 40 s and Task 2 in 50 s. User 2 completed Task 1 in 60 s but did not complete Task 2 successfully, even though 40 s were spent. Applying the formula yields TBE = 0.0154. This value represents the average efficiency in terms of successful tasks per second.

3. Average Time per Task

This metric indicates the average time users spend completing each task and serves as an auxiliary measure for evaluating efficiency.

Average time = Σ time per user / Total number of users

The lower the average time, without compromising the success rate, the better the system's perceived performance.

Example: With 15 tasks performed and a total execution time of 780 seconds, the average time is 780 / 15 = 52 s. This value should be compared against the baseline or reference time.

4. Error Rate

This metric measures how often users make errors while executing tasks. An error may be a wrong click, entering incorrect information in a field, or interrupting the flow of a task.

Error rate = Total number of errors committed / Total number of interactions

This metric helps identify interface points that cause confusion, require additional training, or need redesign.

Example: Suppose users made 6 errors during 15 tasks. Error rate = 6 / 15 = 0.4 [errors per task]. Depending on task criticality, a rate of 0.4 may be considered high and should motivate investigation of interface problem areas.

5. Learning Rate

The learning rate measures the evolution of user performance when a similar task is repeated. In EzList, this analysis was applied to Tasks 1, 2, and 6 (patient insertion), which have an analogous structure.

Learning rate = [(Time T1 + Time T2) − Time T6] / Time (T1 + T2) × 100

- Task 1: enter a patient;
- Task 2: associate a procedure;
- Task 6: enter another patient and associate the patient with a procedure.

If the value of this metric is below 1, it indicates performance gains through familiarization, evidencing a positive learning curve.

Example: If the average time for Task 1 + Task 2 was 60 seconds and the average time for Task 6 was 40 seconds, then the learning rate is 40 / 60 × 100 ≈ 67%. In other words, there was a 33% time gain, suggesting that the system offers a fast and intuitive learning curve.

6. Percentage of Help Requested or Required (via Video or Observation)

This metric quantifies the proportion of users who requested or needed external help while performing tasks in the system, which is useful for evaluating interface self-sufficiency. Help may be classified as:

- Requested help: when the user verbally or gesturally expresses a doubt;
- Required help (not requested): when the evaluator intervenes due to stagnation, prolonged hesitation, or repeated error.

This metric reflects the degree of user dependence on external support and is especially useful in observational tests with video recordings.

$PA = [\Sigma_i \Sigma_j a_{ij} / (U \times T)] \times 100$

- PA: percentage of requested or required help (%);
- a_{ij} = 1 if user i requested or needed help in task j; 0 otherwise.

7. Optimal Paths

An optimal path aims to measure the proportion of users who are able to perform a given task following the expected or ideal flow, defined beforehand based on criteria such as execution time, minimum number of actions, or completion without detours. This indicator reflects the expected efficiency of the system under normal conditions of use and serves as an indicator of interface clarity and fluency.

According to ISO/IEC 25062 (ISO, 2006), the optimal path can be defined as a baseline reference internally established to judge future user performance or future versions of the system. In the case of EzList, the optimal path can be based on the minimum estimated time for each task.

Example: Minimum-time path. It may be estimated by a specialist executing the tasks and/or by the average of the 3 users with the shortest completion times. For Task 2, for example, the minimum observed time among the most fluent users was 85 seconds. Therefore, times ≤ 90 seconds can be considered optimal paths.

Calculations

1. Task Completion Rate

Overall rate: 95.71%

Resident	Rate

P01	85.714286
P02	100.00
P03	100.00
P04	100.00
P05	100.00
P06	100.00
P07	100.00
P08	100.00
P09	100.00
P10	71.43

Command-line execution:

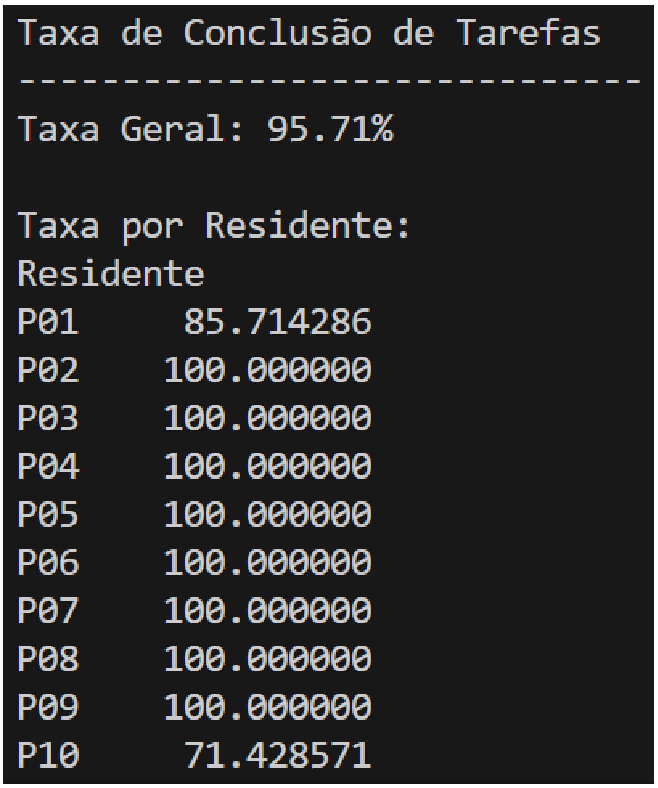


Figure 1 – Execution of Metric 1 via Project EzList_IHC

Source: prepared by the author

2. Time-Based Efficiency

The interpretation of a “good” value for this metric depends on the context of the system, the complexity of the tasks, and usability expectations. In this study, the TBE was:

- TBE (decimal value): 0.0147
- TBE (percentage): 1.47%
- R (number of residents): 10
- N (number of tasks): 7
- Sum of ratios (n_{ij} / t_{ij}): 1.029

Comparison with optimal paths: there is no universal “ideal” TBE value; it must be interpreted within the context of the system. A higher TBE indicates that, on average, tasks are completed successfully in less time. A lower value may indicate that users are spending too much time or failing more frequently.

The optimal paths used are described in item 7.

Command-line execution:

```
=== Debug / Valores Intermediários ===
R (número de residentes): 10
N (número de tarefas): 7

--- Detalhes por (Residente, Tarefa) ---
   Residente    Tarefa  n_ij   t_ij  ratio_ij
0        p01  tarefa 1     0  161.0  0.000000
1        p01  tarefa 2     1  177.0  0.005650
2        p01  tarefa 3     1  109.0  0.009174
3        p01  tarefa 4     1   35.0  0.028571
4        p01  tarefa 5     1   24.0  0.041667
..       ...       ...   ...    ...       ...
65       p10  tarefa 3     1  121.0  0.008264
66       p10  tarefa 4     1   82.0  0.012195
67       p10  tarefa 5     1   31.0  0.032258
68       p10  tarefa 6     0  101.0  0.000000
69       p10  tarefa 7     1  105.0  0.009524
69       p10  tarefa 7     1  105.0  0.009524

[70 rows x 5 columns]
[70 rows x 5 columns]

Soma das razões (n_ij / t_ij) = 1.0295
Denominador (R*N) = 10 * 7 = 70
========================================

Eficiência Baseada no Tempo (TBE) - Média das razões n_ij / t_ij
----------------------------------------------------------------
TBE (valor decimal) = 0.0147
TBE (percentual)    = 1.47%
```

Figure 2 – Execution of Metric 2 via Project EzList_IHC

Source: prepared by the author

```
--- Comparação entre TBE real e TBE ótima ---
TBE real (decimal): 0.0147
TBE real (%): 1.47%
TBE ótima (decimal): 0.0178
TBE ótima (%): 1.78%
Diferença absoluta: 0.30%
Soma das razões n_ij / t_ij (real): 1.0295
Denominador (R*N): 10 * 7 = 70
------------------------------------------------
```

Figure 3 – Execution of Metric 2 with Optimal Paths via Project EzList_IHC

Source: prepared by the author

3. Average Time per Task

Overall average time: 85.31 s

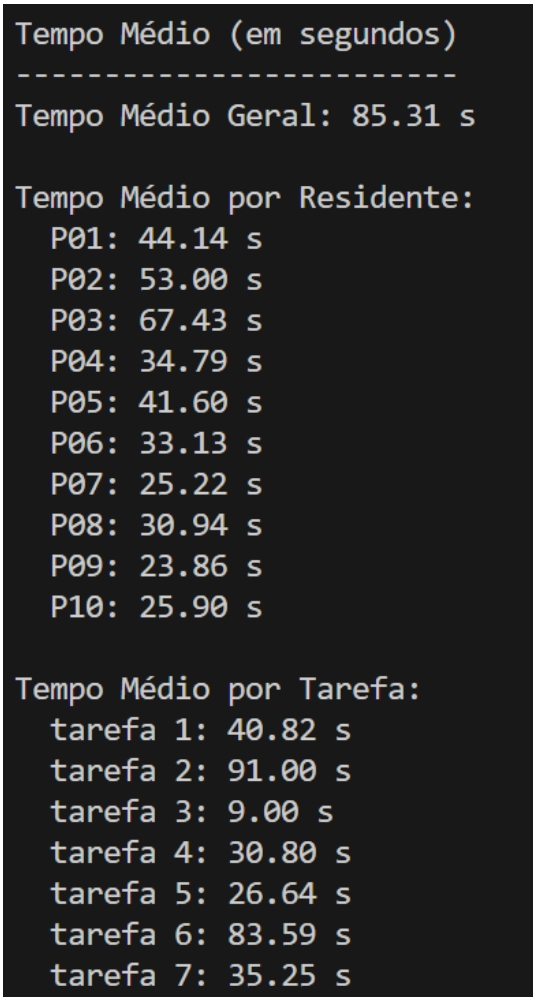
Resident	Average Time (s)
P01	44.14
P02	53.00

P03	67.43
P04	34.79
P05	41.60
P06	33.13
P07	25.22
P08	30.94
P09	23.86
P10	25.90

Average time per task:

Task	Average Time (s)
Task 1	40.82
Task 2	91.00
Task 3	9.00
Task 4	30.80
Task 5	26.64
Task 6	83.59
Task 7	35.25

Command-line execution:



```
Tempo Médio (em segundos)
-------------------------
Tempo Médio Geral: 85.31 s

Tempo Médio por Residente:
  P01: 44.14 s
  P02: 53.00 s
  P03: 67.43 s
  P04: 34.79 s
  P05: 41.60 s
  P06: 33.13 s
  P07: 25.22 s
  P08: 30.94 s
  P09: 23.86 s
  P10: 25.90 s

Tempo Médio por Tarefa:
  tarefa 1: 40.82 s
  tarefa 2: 91.00 s
  tarefa 3: 9.00 s
  tarefa 4: 30.80 s
  tarefa 5: 26.64 s
  tarefa 6: 83.59 s
  tarefa 7: 35.25 s
```

Figure 4 – Execution of Metric 3 via Project EzList_IHC

Source: prepared by the author

4. Error Rate

Global error rate: 27.88%

Resident	Error Rate
P01	7.14%
P02	16.67%
P03	21.43%
P04	31.58%
P05	26.67%
P06	13.33%
P07	33.33%
P08	18.75%
P09	40.91%
P10	50.00%

Command-line execution:

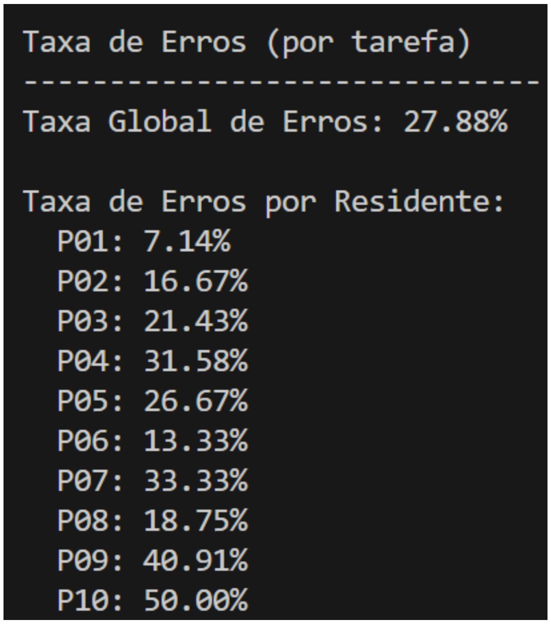


Figure 5 – Execution of Metric 4 via Project EzList_IHC

Source: prepared by the author

5. Learning Rate

Overall learning rate: 16.34%

Resident	Learning Rate
P01	24.56%
P02	30.07%
P03	-3.83%
P04	55.12%
P05	27.27%
P06	-128.15%
P07	-26.38%
P08	38.15%
P09	69.47%
P10	77.10%

Command-line execution:

```
Taxa de Aprendizagem entre Tarefa 1 + 2 e Tarefa 6
--------------------------------------------------
Taxa de Aprendizagem Geral: 16.34%

  P01: 24.56%
  P02: 30.07%
  P03: -3.83%
  P04: 55.12%
  P05: 27.27%
  P06: -128.15%
  P07: -26.38%
  P08: 38.15%
  P09: 69.47%
  P10: 77.10%
```

Figure 6 – Execution of Metric 5 via Project EzList_IHC

Source: prepared by the author

6. Percentage of Help

For this calculation, the markings recorded through direct observation or videos were considered, quantifying the number of executions in which at least one request for help or intervention was recorded. Based on the collected data, the calculated value was 14.29%, indicating that in approximately one out of every seven executions, users depended on assistance to complete the task.

Command-line execution:

```
Percentual de Ajuda Solicitada ou Necessária
------------------------------
PA: 14.29%
------------------------------
```

Figure 7 – Execution of Metric 6 via Project EzList_IHC

Source: prepared by the author

7. Optimal Paths

The construction of the optimal TBE was based on a methodology that combined expert references and empirical data. Specifically, the optimal times per task were defined from the average between the values estimated by two usability specialists and the actual times of the three users with the best performance (shortest time). This mixed approach sought to balance technically grounded judgment with practical observations, thus providing a reasonable benchmark for comparison.

Metric	Value (%)
Current TBE	1.47%
Optimal TBE (baseline)	1.78%

Charts

Frequency of Problem Types

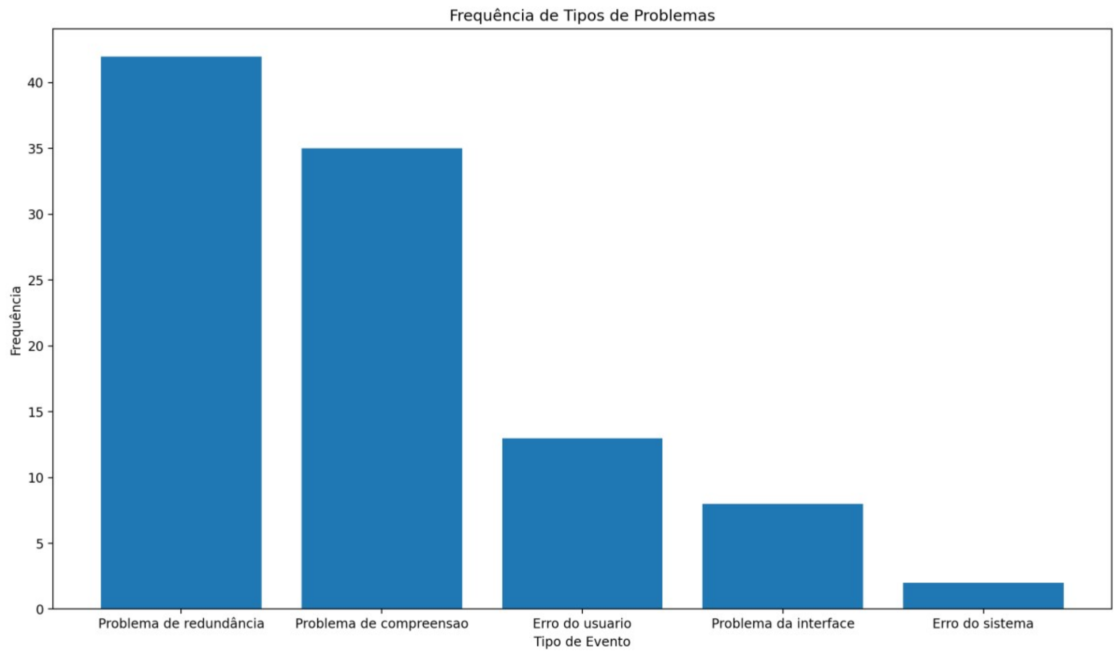


Figure 8 – Chart of Problem Types

Source: prepared by the author

Code call:

python -m gerar_graficos.main --csv C:/caminho_csv/Linha_do_Tempo_EzList.csv --grafico barras

X-axis: each category represents a type of problem (for example, “redundancy problem”, “user error”, and so on).

Y-axis: indicates how many times that problem occurred (frequency).

Description: the taller the bar, the greater the number of occurrences of that problem type.

Total duration by Resident and Event

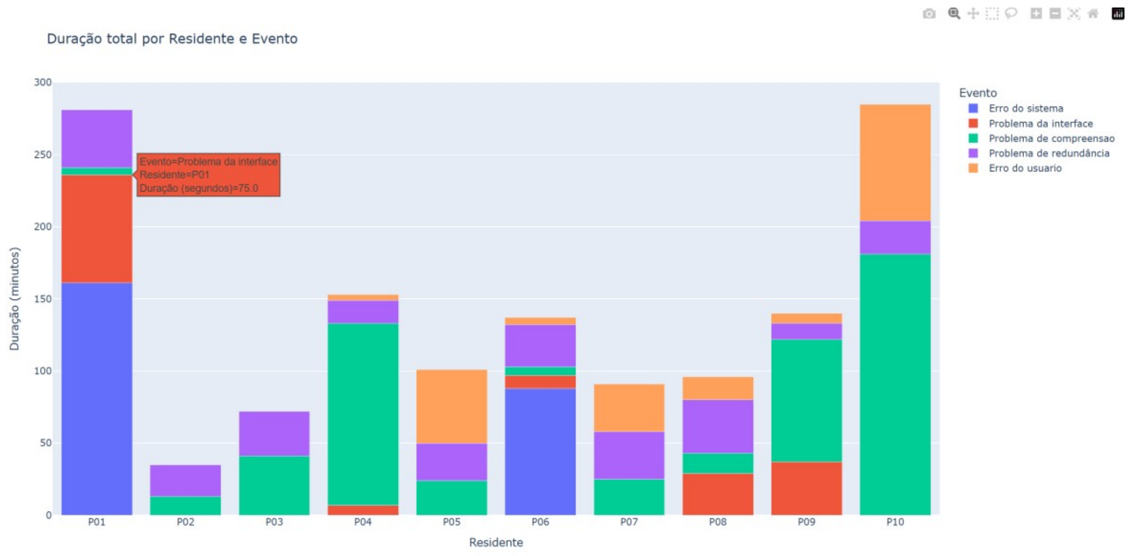


Figure 9 – Total duration by Resident and Event chart

Source: prepared by the author

Code call:

```
python -m gerar_graficos.main --csv C:/caminho_csv/Linha_do_Tempo_EzList.csv --grafico duration
```

X-axis: each resident (P01, P02, …).

Y-axis: total time (in seconds).

Description: each color corresponds to a type of event. The height of each stacked segment is the time spent on that specific event, and the sum of all segments gives the total time the resident spent dealing with problems.

Conclusion

Based on the results obtained, it is possible to form a more precise view of the usability of the EzList system. The task completion rate was high (95.71% overall), indicating that most users were able to finish the proposed activities. However, some residents showed lower performance, such as P10, with a completion rate of 71.43%, which suggests the need for specific adjustments in the flow to ensure greater uniformity.

In this study, the actual TBE obtained was 1.47%, while the optimal TBE (baseline), calculated from estimated ideal execution times, was 1.78%. The difference of 0.30 percentage points below the ideal represents a relative drop in performance that, although numerically modest, indicates that users are performing the tasks less efficiently than expected. This suggests that, although the system is functional, it still imposes barriers or friction that affect the time needed to complete activities successfully.

The definition of the optimal TBE used a hybrid methodology, integrating the evaluation of two specialists with observational data. The ideal times for each task were obtained from the average between the estimates provided by two usability specialists and the times recorded for the three users with the best performance. This strategy sought to reconcile technical analysis with empirical evidence, resulting in a coherent reference parameter for comparison. The fact that the actual TBE is below the ideal reference reinforces the need to investigate points of friction in the interaction flow. Even when users complete tasks successfully, the additional time required may indicate usability aspects that can still be optimized, such as the clarity of instructions, the arrangement of interface elements, or redundant steps.

Regarding error rate, the global mean was 27.88%, with significant variation across residents (from 7.14% to 50%). This indicates that, although the system allows tasks to be completed, some interactions remain prone to failure, as shown by the predominance of “comprehension problems” and “redundancy problems” in the frequency charts.

The learning-rate analysis also shows positive signs, with an overall performance gain of 16.34% between Tasks 1 and 2 and Task 6, which has a similar structure. However, some users showed marked improvement, such as P10 (77.10%) and P09 (69.47%), while others showed a drop in performance, which may indicate that learning is not uniform and may depend on user profile or task complexity.

Average time per task also varied considerably, especially for Task 2, which required more time (91 seconds), suggesting that it is one of the most complex tasks. Given the nature of the task and the user profile (medical residents), however, this was also one of the activities participants performed with greater attention. Task 6 required 83.59 seconds and, because it represented the second execution of the same functionality previously presented in Tasks 1 and 2, it appears to be one of the least intuitive tasks. This information can guide specific interface or flow revisions.

The Percentage of Help Requested or Required (PA) value of 14.29% suggests that, although most users were able to perform tasks autonomously, there are still points in the system that require external intervention, whether due to doubts, confusion, or the complexity of certain steps. Even though this rate is not excessively high, it indicates opportunities to improve interface clarity and interaction fluency, especially to reduce dependence on support during use.

Finally, the graphical analysis of total duration by event confirms that error types such as “comprehension problem” and “redundancy problem” consume a considerable amount of users’ time, reinforcing the importance of reviewing nomenclature, visual feedback, and step structure.

Taken together, these metrics show clear opportunities for improvement in terms of interface clarity, reduction of redundancies, and more uniform performance across users. The adoption of recognized standards, such as ISO 9241-11, helps keep the evaluation aligned with expected parameters for systems in healthcare, promoting a smoother experience for the professionals involved.

References

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